

Multimodal AI Healthcare Triage, Emergency Detection, and Predictive Decision Support Assistant

This project proposes a Multimodal AI-based Healthcare Triage and Decision Support Assistant designed to provide early health assessment by analyzing patient inputs from multiple sources such as text descriptions of symptoms, voice-based symptom reporting, and medical images including X-rays, skin images, injury photos, and health reports. The system integrates advanced AI techniques such as natural language processing for understanding symptom descriptions, speech recognition for voice input processing, computer vision for medical image analysis, classification models for emergency detection and risk categorization, and regression models for predictive healthcare analytics.

Based on the analyzed inputs, the assistant identifies potential emergency conditions, classifies patient risk levels into categories such as Low, Moderate, High, and Critical, and recommends appropriate next actions such as monitoring symptoms, consulting a doctor, or seeking immediate medical attention. In addition to classification-based decision support, the system also performs predictive analysis by estimating patient risk scores, expected hospital waiting time, and approximate recovery duration using regression techniques. A health dashboard is included to present symptom history, risk assessments, emergency alerts, predictive insights, and recommended actions in an organized and user-friendly manner. The project is intended as an AI-powered healthcare support system to assist in early triage, prioritization, and informed decision-making, while not replacing professional medical diagnosis or prescribing medications.